

Lema de Borel-Cantelli I

Lema 1 (de Borel-Cantelli I). Sea $(A_n)_{n \in \mathbb{N}} \subset \Sigma$ en $(\Omega, \Sigma, \mathbb{P})$ tal que $\sum_n \mathbb{P}(A_n) < \infty$

$$\implies \mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) = 0.$$

Demostración: Definimos $N: \Omega \rightarrow \mathbb{R} \cup \{\infty\}$ dada por $N(\omega) = \sum_{n \in \mathbb{N}} \mathbb{1}_{A_n}(\omega)$, entonces

$$\limsup_{n \rightarrow \infty} A_n = \{\omega \in \Omega : N(\omega) = \infty\}.$$

Por tanto, $\mathbb{E}[N] \stackrel{\text{TCM}}{=} \sum_{n \in \mathbb{N}} \mathbb{E}[\mathbb{1}_{A_n}] = \sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty.$

$$\implies N \in \mathcal{L}^1(\mathbb{P}) \implies \mathbb{P}(\{\omega \in \Omega : N(\omega) = \infty\}) = 0.$$

Concluimos por tanto que $\mathbb{P}(\limsup A_n) = 0$. ■

Referenciado en

- ley-fuerte-grandes-numeros
- prop-borel-cantelli-iii
- teo-convergencia-martingalas
- convergencia-probabilidad-imp-sucesion-casi-segura

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.